Artificial Morphogenesis using Evolutionary Cellular Automata

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Motivation

Fundamental Inquiry: Can simple, discrete systems generate complex emergent structures?

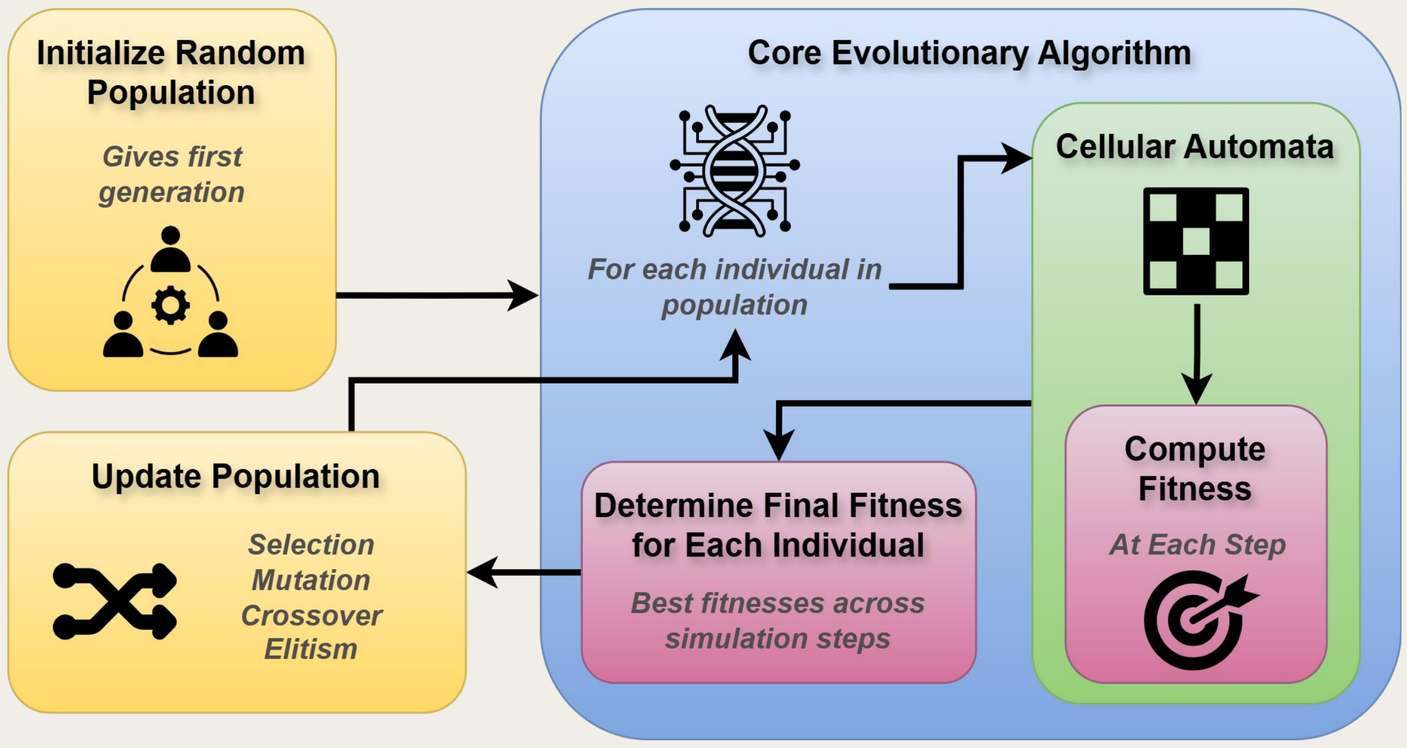
Interpretability Gap: While neural CA are effective, their black-box nature limits structural analysis.

Proposed Approach: We investigate the evolution of fully discrete CA rule sets for targeted morphogenesis.

RQ 1 How does fitness design affect convergence?

RQ 2 Do higher fitness scores indicate better visuals?

RQ 3 Is there an optimal number of CA steps for stable pattern formation?

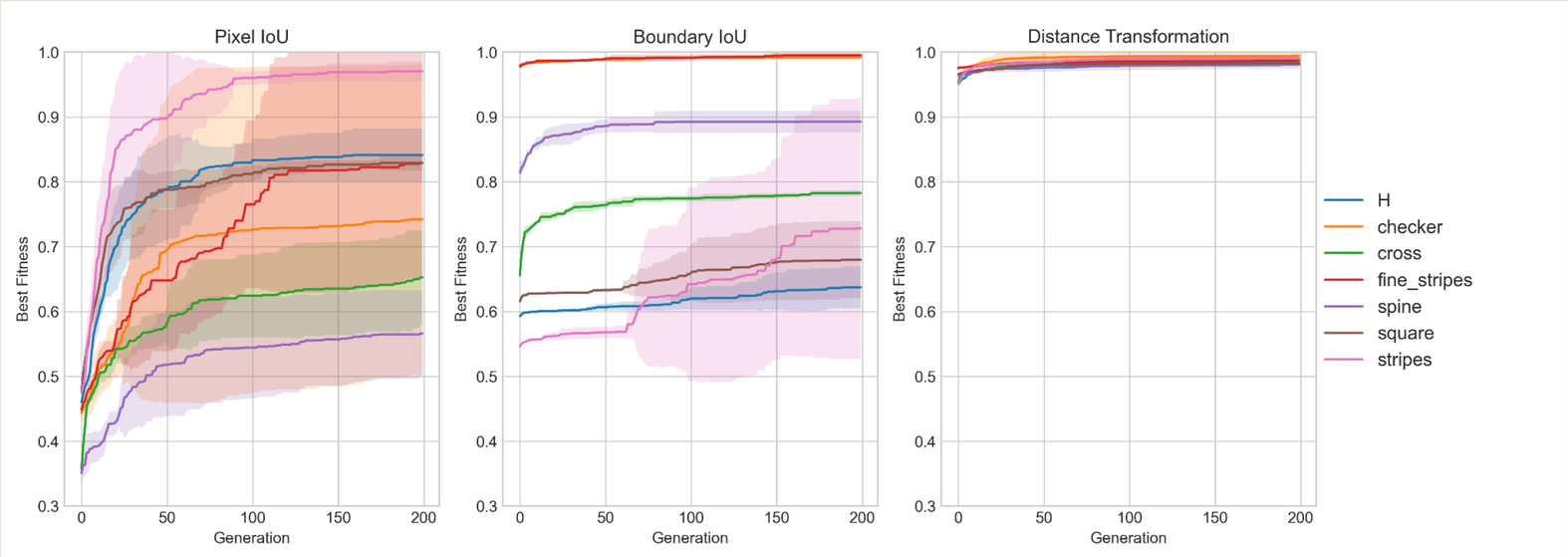


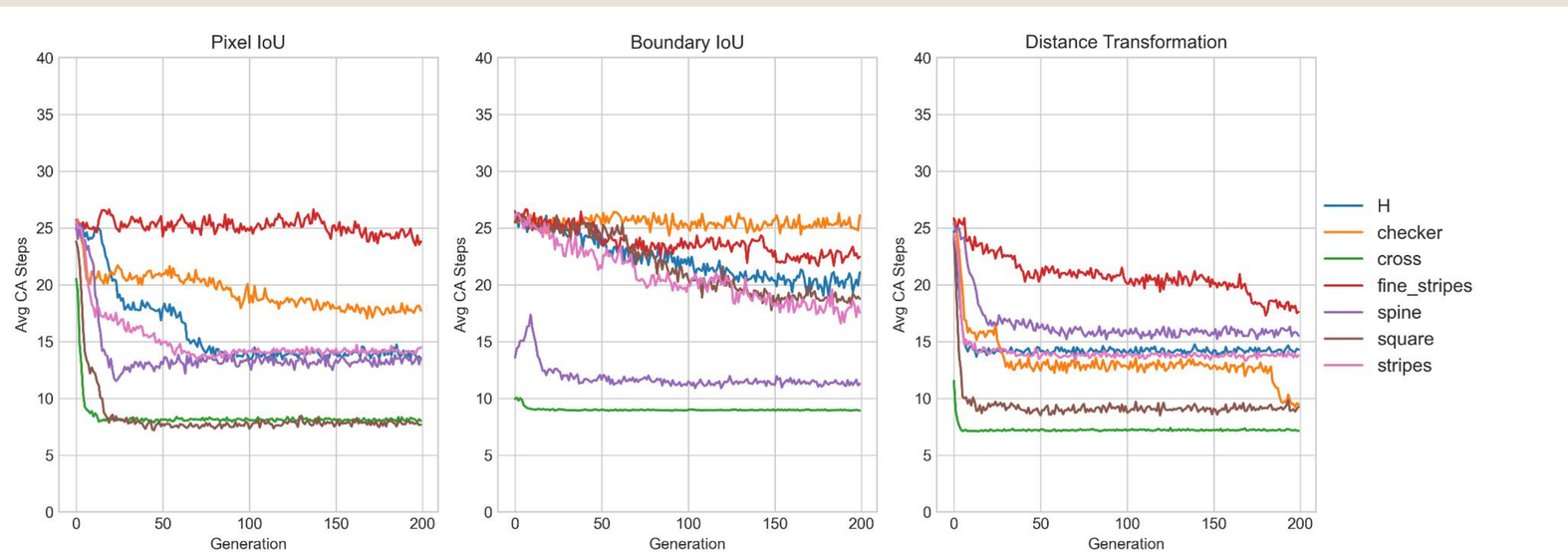
Fitness Metrics and Results

Pixel IoU
Measures pixel overlap

Boundary IoU
Prioritizes edge similarity

Distance Transform (DT)
Captures global structure





Pattern	Best	Target	Fitness Metric
Checker			DT (Tied Pixel IOU)
Cross			DT
H			Pixel IOU
Fine Stripe			DT (Tied Pixel IOU)
Spine			Pixel IOU
Square			Boundary IOU
Stripe			Pixel IOU

Conclusions

- Pixel IoU is most effective for fine structures
- Boundary IoU captures large, simple shapes
- Distance Transform similarity enables the fastest convergence via global structural alignment
- The "sweet spot" for stable pattern growth is 7–25 CA iterations
- High fitness values can be misleading and do not guarantee visual similarity
- Proper objective alignment is critical; the algorithm will optimize exactly what is measured

Future Work

- Multi-objective fitness
- Long-term stability
- Experimental neighborhoods
- Complex target pattern design

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'artificial-morphogenesis-evo'
On GitHub

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